



Adolescent smartphone use, sleep, and physical activity: daily associations between sensor-based measures in the adolescent brain cognitive development (ABCD) study

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ABSTRACT

Purpose: Potential health consequences of adolescent smartphone use are a growing public concern. Improving upon existing, largely self-report-based research, this study investigated relationships between adolescent smartphone use, sleep, and physical activity using passive sensor measures.

Methods: Over three weeks, 791 Adolescent Brain Cognitive Development (ABCD) Study participants ($M_{age}=14.12$, 53 % female) provided smartphone application and keyboard use data via the Effortless Assessment Research System (EARS) application, and sleep and physical activity data via Fitbit device.

Results: Mixed-effects models found that daytime/evening application use (6:00AM–7:59PM) predicted reduced exercise, more time sedentary, and fewer daily steps (Standardized $\beta=-0.21-0.07$, $P<.001$). Late-evening use (8:00PM–9:59PM) modestly predicted increased sedentary time and reduced steps (Standardized $\beta=-0.05-0.02$, $P<.001$). Late-night use (10:00PM–5:59AM) predicted reduced sleep, delayed sleep onset, delayed waking, increased sedentary time, and fewer daily steps (Standardized $\beta=-0.16-0.27$, $P<.001$). After disaggregating within and between-person smartphone use, within-person relationships remained significant, with associations of similar magnitude to the initial analyses (daytime/evening use: standardized $\beta=-0.22-0.07$, $P<.001$; late-evening use: standardized $\beta=-0.05-0.02$, $P<.001$; late-night use: standardized $\beta=-0.16-0.24$, $P\leq.002$), indicating daily-level relationships unattributable to between-subject differences. Examining smartphone use effects by hour relative to sleep onset indicated that only use recorded after initial sleep onset significantly predicted sleep, while use recorded 3–12 hours before sleep onset significantly predicted step counts.

Conclusions: Using passive sensor data, we found significant associations between adolescent smartphone use, physical activity, and sleep which differed by time of day and remained significant within subjects. Experimental replication is recommended to strengthen tentative causal claims.

1. Introduction

US adolescents spend an estimated 5.6 hours per day using their smartphones, with many recording 10+ hours of daily use (Christakis et al., 2025). This extensive use represents a growing concern for parents, policymakers, and health care providers, who worry affected adolescents are at increased risk of physical and mental health consequences.

Two areas of especially acute concern for smartphone related consequences are reductions in sleep and physical activity. Inadequate sleep in adolescence has been linked to increased mood, somatic, and behavioral problems as well as impaired school performance (Shochat et al., 2014) while physical inactivity in adolescence is associated with obesity, depression, and increased cardiovascular and metabolic disease risk (Sluijs et al., 2021).

Recent systematic reviews support associations between extensive

Abbreviations: ABCD Study, Adolescent Brain Cognitive Development Study; EARS App, Effortless Assessment of Risk States Smartphone Application; OS, Operating System; App, Smartphone Application; METs, metabolic equivalents.

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smartphone use, sleep, and physical activity in adolescence. Higher self-reported nighttime smartphone use has been linked to reduced sleep duration, poorer sleep quality, delayed sleep onset, and increased daytime tiredness in adolescents (Brautsch et al., 2023; de Sá et al., 2023). A meta-analysis of randomized, controlled trials of childhood sleep interventions targeting smartphone use identified intervention effects including reduced screen time, increased total sleep, and earlier bedtimes, though its interpretation is complicated by heterogeneous study designs and target population ages (which ranged from 2–14 years old), the use of predominantly self and parent-report based measures of screentime and sleep, and a lack of studies which directly experimentally manipulated screentime (Martin et al., 2021). Similarly, a recent systematic review and meta-analysis reported a small-to-moderate effect of self-reported problematic smartphone use on adolescent physical activity, including increased sedentary behavior and reduced exercise (Xiao et al., 2022). The few studies that use methods other than self-report (e.g., video observation; digital trace) provide evidence of nuanced patterns of smartphone and screen use which provide unique insights into adolescent sleep and behavior (Brosnan et al., 2024; Lee et al., 2017; Tkaczyk et al., 2024).

Hypothesized causal explanations for links between smartphone use and poorer sleep include circadian rhythm disruptions due to late-night blue light exposure, sleep disruption due to being woken by late-night device notifications, and difficulty disengaging from energizing content before bed (Hartstein et al., 2024; LeBourgeois et al., 2017; Rideout et al., 2022). More recently, research has posited causal effects in the opposite direction, with adolescents using digital technologies to fill time during sleep delays or to regulate negative emotional states caused by sleep difficulties (Bauducco et al., 2024). Time displacement, where smartphone use replaces time spent physically active, is most frequently invoked to explain associations between smartphone use and physical activity, (Epstein et al., 2005; Hands et al., 2011; Sandercock et al., 2012; Xiao et al., 2022), as well as sleep (Bauducco et al., 2024; Hökby et al., 2025), though more speculative alternative mechanisms, including screentime-induced executive functioning deficits (Li et al., 2022; Marciano et al., 2021) and screentime impacts on depression and behavioral deactivation have also been proposed to explain these relationships (Liu et al., 2016).

Such interpretations are complicated by widely noted measurement and study design concerns (Brautsch et al., 2023; Xiao et al., 2022). Nearly all studies identified in recent reviews of adolescent smartphone use, sleep, and physical activity exclusively used self-report assessments, often at only a single measurement occasion (Brautsch et al., 2023; Mac Cárthaigh et al., 2020; Xiao et al., 2022). Self-report-based designs are frequently subject to reporting and recall biases (Paulus and Vazire, 2007). Indeed, self-report measures of smartphone use, (Alexander et al., 2024; Harris et al., 2017) physical activity, (Doma et al., 2017; Healey et al., 2020; Prince et al., 2008) and sleep duration (Erwin and Bashore, 2017; Lauderdale et al., 2008) often exhibit poor psychometric properties and show substantial discrepancies from objective, sensor-based measures.

Passive sensor measures of these behaviors may offer important advantages over self-report (Alexander et al., 2023; Brautsch et al., 2023; Xiao et al., 2022). Mobile or wearable device sensors passively record data from participants' real-world environments, improving ecological validity and reducing self-report biases (Alexander et al., 2025; Harari et al., 2016). Commercially available smartwatches, such as the Fitbit Charge2, are acceptable to adolescent research participants (Wing et al., 2022), and offer reasonably accurate measures of sleep and aspects of physical activity (de Zambotti et al., 2018; Germini et al., 2022; Godino et al., 2020; Kiss et al., 2025; Wing et al., 2022). Similarly, smartphone-recorded measures of device use, such as time logs of application and keyboard use, offer reliable, informative measures of smartphone use (Alexander et al., 2024; Ryding and Kuss, 2020; Wade et al., 2021).

The intensive longitudinal data collected by passive sensors may be

especially useful for investigating purported causal effects of smartphone use because they allow for the disaggregation of within-person and between-person effects (Shiffman et al., 2008). Within-person designs measure relationships between two constructs across multiple observations of the same individual, controlling for differences in sociodemographic features, longer-timescale environmental experiences, and genetics that may otherwise confound relationships between behaviors. These designs thereby offer stronger tests of putative causal associations than in observational studies that examine mean differences between subjects (Lu, 2009). For instance, Burnell et al. (2022) challenged the purported causal link between adolescent smartphone use and sleep by finding that adolescent smartphone use and sleep were significantly related only between subjects: suggesting that day-to-day changes in smartphone use were unrelated to changes in sleep (Burnell et al., 2022). Passive measurement strategies thus represent a feasible approach to reduce reporting biases, increase ecological validity, and improve causal inference in studies of smartphone use and adolescent health.

The present study employed continuous, passive sensor data to more richly and precisely measure relationships between adolescent smartphone use, sleep, and physical activity. Data were collected during a three-week trial within the Adolescent Brain Cognitive Development (ABCD) Study. This study had three primary aims. First, we assessed whether prior cross-sectional, self-report-based associations between adolescent smartphone use, sleep, and physical activity were replicated using sensor-based measures in intensive longitudinal data. Second, we investigated whether relationships differed by time of day (i.e., between daytime or evening smartphone use and late-night smartphone use). Third, we disaggregated between-person and within-person effects of smartphone use, examining the extent to which associations were explained by individuals' average smartphone use or by daily variation in their use. In doing so, the present study aimed to provide further evidence on possible impacts of smartphone use on sleep and physical activity in adolescence, addressing many of the methodological limitations of prior research into these behaviors.

2. Methods

2.1. Participants

Data were drawn from the Year-4 (ages 13–14) follow-up visit of the Adolescent Brain Cognitive Development (ABCD) Study, a longitudinal study of child and adolescent development undertaken at 21 research sites nationwide. Data release 5.1 (doi: 10.15154/z563-zd24) was used, which contains data from over half the cohort at Year 4 follow-up. The present study considers the 791 ABCD participants who, following their Year-4 follow-up visit, concurrently provided up to three weeks of passively sensed sleep, physical activity and smartphone use data (see **Supplementary Figure S1**, for additional information on sample inclusion and attrition). **Table 1** reports participant sociodemographic information. A prior investigation of participants in the ABCD EARS trial revealed that, compared to the broader US population of 10–14 year olds at the time of assessment, the sample included significantly more female and non-Hispanic white participants and significantly fewer participants from low-income families (i.e., a combined household income below \$50,000) or families where neither parent had attended college (Alexander et al., 2024).

2.2. Procedures

Study procedures were approved by a central Institutional Review Board at the University of California San Diego (Barch et al., 2018). Full ABCD study procedures are described extensively in prior literature (Barch et al., 2018; Garavan et al., 2018). During the Year-4 follow-up assessment, ABCD participants were asked to provide passive sensor data on their smartphone use, sleep, and physical activity concurrently

Table 1
Participant Demographics.

Measure	N (%)
Sex	
Male	369 (46.6 %)
Female	422 (53.4 %)
Race/Ethnicity	
Asian	17 (2.1 %)
Black	75 (9.5 %)
Hispanic	144 (18.2 %)
White	471 (59.5 %)
Another Race	84 (10.6 %)
Highest Parent Educational Attainment	
No High School Diploma	15 (1.8 %)
High School Diploma/GED	40 (5.1 %)
Some College	241 (30.5 %)
Bachelor's Degree	225 (28.4 %)
Graduate or Professional Degree	270 (34.1 %)
Smartphone OS	
iOS	486 (61.4 %)
Android	305 (38.6 %)
Age at Year-4 Follow Up Assessment: Mean (SD)	14.12 (0.69)
Total N	791 (100.0 %)

for three weeks. Participants installed the Effortless Assessment Research System (EARS) application onto their personal smartphones (Lind et al., 2023). To ensure accurate data collection and syncing, they were instructed to open the app at least once per day and keep it running in the background at all times. Participants were also loaned a Fitbit Charge device and instructed to wear it continuously, except during daily charging. Research assistants monitored data collection and contacted participants to troubleshoot issues (Godino et al., 2020; Wade et al., 2021). Data collection ceased after a participant provided 21 consecutive days of data or after 42 days elapsed. Participants were compensated separately for EARS and Fitbit participation based on the amount of data provided.

The EARS application passively collected smartphone use data continuously for the duration of the study. Due to iOS constraints on third-party apps, the EARS app differed between iOS and Android smartphones. In Android users, the app queried 24-hour smartphone use logs every fifteen minutes, obtaining the foreground application's name, category, and opening and closing timestamps (i.e. when the application was first opened in the device's foreground and when it was closed or minimized or when the screen was turned off). In iOS users, application use time logs of were unavailable due to third party constraints, though comparable information was available via keystroke logs. The keyboard data, which was available for both Android and iOS users, included the name and category of the foreground application and the length of time the keyboard was in use, (i.e. the time between the first and last keystroke recorded during each keyboard session) though keystroke content (e.g., the content of participants' messages) was not retained to maintain participants' privacy. While keyboard measures were identical between Android and iOS users, the data collection method differed across operating systems. For Android users, keyboard use was recorded from the device's native keyboard. For iOS users, it was recorded via a keyboard overlay installed with the EARS application, which recorded participants' keystrokes and which participants were instructed to exclusively use during the study. For the present study, times logged by foreground app usage (for Android) and keyboard logged app usage (for both iOS and Android) are used.

Fitbit devices gathered data using embedded sensors, including a 3-axis accelerometer, heart-rate monitor, and altimeter. These data were processed using proprietary algorithms (unavailable to the study team) to infer behaviors including sleep duration, sleep onset and wake times, step counts, and time spent sedentary or engaged in physical activity. Existing research suggests that Fitbit devices offer reasonably accurate measures of step counts, (Germini et al., 2022; Godino et al., 2020) sleep duration, (Feehan et al., 2018) and, to a lesser extent, physical activity

level (Fuller et al., 2021; Germini et al., 2022; Sushames et al., 2016).

2.3. Measures

2.3.1. Sociodemographic characteristics

Sociodemographics were parent-reported (Barch et al., 2018). Age was the participant's age in months during their Year 4 follow up visit (mean-centered prior to modeling). Sex was sex assigned at birth.

2.3.2. Seasonality

Due to varying school schedules, summer vacation was liberally defined as occurring between June 1st and August 31st.

2.3.3. Weekday/Weekend and school night/ weekend night

"School nights" were Sunday - Thursday nights and "weekend nights" were Friday and Saturday nights. Weekdays were Monday - Friday and weekends were Saturday/Sunday.

2.3.4. Operating system

Smartphone OS were either iOS or Android.

2.3.5. Total fitbit wear time

Total wear time was the time in minutes of recorded fitbit wear time, excluding minutes excluded during quality control procedures.

2.3.6. Sleep measures

Sleep measures were calculated from Fitbit-derived nightly sleep stage data recorded at 30-second intervals. Quality control steps included removing observations with heart rate values below 50 bpm and strings of observations with >10 consecutive identical heart rates (Wing et al., 2022).

Total sleep was defined as the total minutes recorded asleep, regardless of sleep stage, between 5:00PM and 4:59PM the following day. As adolescents frequently exhibit delayed sleep onset and wake times, (Tarokh et al., 2019) this broad sleep window was chosen to prevent underestimation when sleep extended into the following afternoon. Sleep onset was the first moment recorded asleep each night. Wake times were the first moment recorded awake the following morning followed by at least 30 consecutive minutes awake (to prevent misidentification of brief nighttime awakenings as waking).

2.3.7. Daily step count

Daily step counts were the number of Fitbit-recorded steps each day. Calculated steps excluded recordings with missing heart rate values, observations with heart rates below 50 beats per minute, and strings of 10 or more consecutive reads with identical heart rates (Wing et al., 2022). Days with fewer than 599 min of valid waking wear time, days where >20 % of daily steps were excluded during quality control procedures, and days without at least 30 s of recorded sleep were also excluded from analyses.

2.3.8. Sedentary behavior/ physical activity

Minutes of sedentary behavior and moderate to vigorous exercise were measured using estimated metabolic equivalents (METs), the ratio of energy expenditure during an activity vs. at rest. Minutes of sedentary activity were the number of minutes each day with METs <1.5 while minutes of moderate-to-vigorous exercise were the daily number of minutes with METs ≥ 3.0 .

2.3.9. Smartphone application use

Smartphone application use (available for Android participants only) was defined as the amount of time when any smartphone application, including both third party and native applications, was foregrounded in the participant's screen each day. System applications and applications intended to remain in the foreground while participants slept (e.g. lock screen displays or alarm clock apps) were removed prior calculation.

Smartphone use recorded simultaneously with sleep was observed infrequently, on 1.3 % of observed days, though such observations were retained as it was not possible to determine whether this represented real use, like leaving a device on while falling asleep, or was artefactual. Smartphone application use was separated into “daytime/evening” use, recorded between 6:00AM and 7:59PM, “night time” use, recorded between 8:00PM and 9:59PM, and “late-night” use, recorded between 10:00PM and 5:59AM the following morning (corresponding to an expected school night sleep window).

2.3.10. Keyboard use

Keyboard use, a measure of active smartphone use, was measured by the total duration when a participant’s keyboard was open and logging keystrokes. As with smartphone application use, minutes of keyboard use were separated into “daytime/evening” use, recorded between 6:00AM and 7:59PM, “night time” use, recorded between 8:00PM and 9:59PM, and “late-night” use, recorded between 10:00PM and 5:59AM the following morning. To maintain consistency with smartphone application use quality control procedures, keystrokes associated with system applications or applications intended to be foregrounded during sleep were also excluded from keyboard use measures.

2.4. Analytic strategy

Relationships were modeled via mixed effects models using the *nlme* package in R version 4.3.0 (Pinheiro et al., 2021). Each model estimated the effect of one of six smartphone use measures, with same-day total step counts, minutes of sedentary behavior, and minutes of moderate to vigorous exercise, same-night time of sleep initiation and minutes of total sleep, and next-morning wake time. All pairwise complete observations (i.e., all days where both the model outcome and smartphone use measure were available) were included for each model. Each model included fixed and individual-level random intercepts, fixed effects of smartphone use, measured by one of four smartphone use measures (minutes of daytime/evening or late-night keyboard or application use). Models also included sex, age, weekend/weekday, summer vacation status, and total fitbit wear time as covariates. Models estimating the effect of keyboard use included both Android and Apple iOS users and thus included smartphone operating system as an additional covariate.

Models disaggregating within-person and between-person effects of smartphone use included a between-person effect, the participants’ mean smartphone use, and a within-person effect, a participant-mean-centered measure of daily smartphone use computed by subtracting participants’ mean smartphone use from their daily smartphone use.

Models were subjected to an extensive set of sensitivity analyses. Uncorrected models (i.e., those where smartphone use was not decomposed into within/between person effects) were rerun with race and socioeconomic status included as covariates. Models of physical activity outcomes were rerun covarying for total time recorded awake each day. To assess whether “late-night” smartphone use effects differed under alternative definitions of “late-night” use, models estimating late-night application and keyboard use effects were rerun with “late-night” redefined as 9:00PM– 4:59AM and as 11:00PM– 6:59AM. To assess whether results were unduly influenced by outlying sleep or activity values, models were rerun after winsorizing outcome variables to ± 3 SD from the mean (Beaumont and Rivest, 2009). Lastly, to correct for sample attrition, models were rerun after inverse-propensity weighting cases by participants’ proportion of complete days of data, such that observations from participants who provided fewer days of data were assigned additional weight (Seaman and White, 2013).

A final set of follow-up models aimed to estimate smartphone use effects with increased temporal resolution and to investigate whether these effects differed over time relative to participants’ sleep onset. Smartphone application use was binned into 1-hour intervals spanning 12 hours before and after recorded sleep onset. A series of mixed effects regressions were then estimated for both total sleep duration and daily

step count. These models included individual random intercepts fixed effects covariates for sex, age, weekday/weekend status, summer vacation status, and Fitbit wear time, and fixed effects of within-person and between-person smartphone application use. Each model was fit separately for a single time bin, allowing for the assessment of how the magnitude and direction of associations varied across the 24-hour window surrounding sleep. Results were plotted to visualize temporal patterns in smartphone use effects on sleep and physical activity in **Supplementary Figure S1**.

3. Results

Descriptive statistics for each measure of smartphone use, physical activity, and sleep are reported in **Table 2**. In total, eligible participants provided 11,797 days of keyboard use data while eligible Android device users further provided 6340 days of smartphone application use data. Participants further provided 6748 days of concurrent smartphone use and physical activity data and 5634 nights of concurrent smartphone use and sleep data. Keyboard use between 10:00PM and 5:59AM was recorded on 44.1 % of nights, with 83.5 % of participants registering late night use on at least one occasion. Similarly, application use between 10:00PM and 5:59AM on 74.7 % of recorded nights, with 97.6 % of Android users recording at least one instance of late-night use. Further differences in sociodemographic characteristics and patterns of smartphone use between Android and iOS participants have been described previously in Alexander et al. (2024).

Results of mixed effects models testing associations between smartphone use sleep and physical activity are presented in **Table 3**, and in the supplementary material. Smartphone application use recorded between 6:00AM and 7:59PM predicted 7.11 fewer steps, 3 fewer seconds of exercise, and 10 more seconds of sedentary activity per minute of use (standardized $\beta = -0.21 - 0.07$, all $P \leq .001$). Evening smartphone use, recorded from 8:00PM–9:59PM predicted 6.78 fewer steps and 13 additional seconds spent sedentary per minute of use (standardized $\beta = -0.05 - 0.002$, all $P \leq .001$). Late-night smartphone application use, between 10:00PM and 5:59AM the next morning, predicted 3.22 fewer steps taken, 4 more seconds of sedentary activity, 10 fewer seconds of sleep, a 22 s sleep onset delay, and a 15 s delay in waking the next morning per minute of use (standardized $\beta = -0.16 - 0.27$, all $P \leq .001$).

Similarly, each minute of keyboard use recorded between 6:00AM and 7:59PM predicted 5 fewer seconds spent exercising, 21 more sedentary seconds, 12 fewer steps, and a 12 s sleep onset delay (standardized $\beta = -0.05 - 0.03$, all $P \leq .038$). Each additional minute of keyboard use between 8:00PM and 9:59PM predicted a 30 s delay in sleep onset (standardized $\beta = 0.03$, $P = .045$). Lastly, each minute of late-night keyboard use (between 10:00PM and 5:59AM) predicted 42 fewer seconds of sleep, a 93 s delay in sleep onset and a 57 s delay in waking the next morning (standardized $\beta = -0.09 - 0.13$, all $P \leq .001$).

When included as a fixed effects covariate, smartphone operating system was significantly associated with physical activity measures, with iOS users recording more daily steps and more minutes of exercise (standardized $\beta = 0.09 - 0.18$, all $P < .044$) and fewer minutes of sedentary behavior (standardized $\beta = -0.17 - -0.16$, all $P < .007$), though there were no significant effects of OS on sleep outcomes (all $P > .13$). Full model outputs, which include further information on covariate effects, are available in the supplementary material.

Between-person effects, representing a participant’s average daily smartphone use, and within-person effects, representing daily smartphone use above or below the participant’s average use, are reported in **Fig. 1**, **Table 3**, and in the supplementary material. Significant between-subject effects of smartphone and keyboard use were identified for 5/6 sleep and physical activity outcomes (none were identified for time spent exercising) while significant within-subject effects were identified for all six outcomes.

At the between-subject level, smartphone use from 6:00AM–7:59PM predicted significantly later sleep onset, increased sedentary behavior,

Table 2

Descriptive Statistics: Smartphone Use, Sleep, and Physical Activity Measures.

Measure	Grand Mean: Mean (SD)	Participant Mean: Mean (SD)	N_{days} : N (%) missing	$N_{participants}$	Days-per-Participant: Mean (SD)	McDonald's ω
Smartphone Use Measures						
Minutes of Keyboard Use (6:00AM – 7:59PM)	11.68 (21.28)	9.89 (14.62)	11,797 (28.98 %)	791	14.91 (7.23)	0.97
Minutes of Keyboard Use (8:00PM – 9:59PM)	2.90 (7.27)	2.47 (4.52)	11,797 (28.98 %)	791	14.91 (7.23)	0.93
Minutes of Keyboard Use (10:00PM – 5:59AM)	4.04 (12.25)	3.33 (7.68)	10,159 (38.88 %)	791	12.84 (7.38)	0.92
Minutes of Smartphone Use (6:00AM – 7:59PM)	190.47 (140.26)	188.68 (96.00)	6340 (0.03 %)	302	20.99 (3.86)	0.95
Minutes of Smartphone Use (8:00PM – 9:59PM)	37.66 (36.89)	36.98 (23.10)	6340 (0.03 %)	302	20.99 (3.86)	0.93
Minutes of Smartphone Use (10:00PM – 5:59AM)	71.52 (97.11)	70.53 (69.58)	5993 (5.50 %)	302	19.84 (3.97)	0.95
Physical Activity Measures						
Total Daily Steps	7553.73 (4578.02)	7823.82 (3784.77)	6228 (62.5 %)	791	7.87 (6.12)	0.94
Minutes of Sedentary Activity	973.01 (333.41)	935.51 (316.73)	6748 (59.4 %)	791	8.53 (5.92)	0.99
Minutes of Moderate or Vigorous Exercise	31.73 (47.52)	34.57 (36.76)	6748 (59.4 %)	791	8.53 (5.92)	0.93
Sleep Measures						
Total Minutes of Sleep	449.98 (95.39)	441.48 (66.65)	5634 (55.6 %)	604	9.33 (6.48)	0.81
Bed Times	11:48PM (2 h, 14 m)	12:01AM (1 h, 43 m)	5634 (55.6 %)	604	9.33 (6.48)	0.93
Wake Times	8:02AM (2 h, 17 m)	8:04AM (1 h, 38 m)	5375 (53.0 %)	604	9.30 (6.28)	0.90

Grand means and SDs include all observations across all participants and study days. Participant means and SDs are means and SDs of within-subject means. N_{days} are the number of days of data recorded for each measure by eligible participants. Days-per-participant are the mean and standard deviation of the number of days of data recorded per participant for each measure.

and reduced total steps (standardized $\beta = -0.12 - 0.15$, all $P \leq .038$) while, at the within person level, this measure predicted fewer steps, fewer minutes of exercise, and more minutes of sedentary behavior (standardized $\beta = -0.22 - 0.07$, all $P \leq .001$). For smartphone use from 8:00PM – 9:59PM, between subject effects predicted significant sleep reductions and sleep onset delay (standardized $\beta = -0.18 - 0.26$, all $P \leq .001$) while within-subject effects predicted significantly fewer steps and increased sedentary time (standardized $\beta = -0.05 - 0.02$, all $P \leq .002$). Late night smartphone application use (10:00PM – 5:59AM) was associated at the between-subject level with reduced step counts, reduced sleep duration, and delayed sleep onset and morning waking (standardized $\beta = -0.18 - 0.44$, all $P \leq .001$); at the within person level, late night application use predicted increased sedentary behavior, reduced step counts, reduced sleep duration, and sleep onset and morning waking delays (standardized $\beta = -0.16 - 0.24$, all $P \leq .002$).

Comparable results were observed for keyboard use measures. From 6:00AM to 7:59PM, between-person keyboard use predicted increased sedentary time (standardized $\beta = 0.09$, $P = .027$) while within-person keyboard use significantly predicted fewer total steps, fewer minutes of exercise and greater time spent sedentary (standardized $\beta = -0.05 - 0.02$, all $P \leq .001$). From 8:00PM to 9:59PM, keyboard use at the between subject level predicted delayed sleep onset (standardized $\beta = 0.10$, $P \leq .028$) but had no significant within-person effects. Lastly, from 10:00PM to 5:59AM, between-person keyboard use predicted reduced total steps, reduced sleep duration, and delayed sleep onset and morning waking (standardized $\beta = -0.12 - 0.34$, all $P \leq .038$) while, at the within subject level, late-night keyboard use significantly predicted reduced sleep duration and delayed sleep onset and morning waking (standardized $\beta = -0.08 - 0.12$, all $P \leq .001$).

Sensitivity analyses run in addition to the primary models are reported in the supplementary material. Broadly, the magnitudes and patterns of statistical significance for smartphone use effects were highly similar between the primary analyses and sensitivity models. In each sensitivity analysis, standardized smartphone use effects were all within

0.10 SD of their corresponding effects in the primary models. All statistically significant effects identified in the primary models remained statistically significant after correcting for race and socioeconomic status. After covarying for time recorded awake in the physical activity models, 11/15 statistically significant effects remained significant and one additional effect was significant in the sensitivity model but not in the primary model. After shifting the definition of “late-night” use one hour later (to 11:00PM – 6:59AM), patterns of statistical significance were identical to the corresponding primary models. Shifting the definition of “late-night” one hour earlier, to 9:00PM – 4:59AM, attenuated two (out of 31 total) statistically significant effects from the primary models to below statistical significance. After winsorizing outcome values ± 3 SD from the mean, 30/31 statistically significant effects in the primary models remained statistically significant in the sensitivity analysis. Inverse-propensity weighting data based on participants’ proportion of non-missing data reproduced 25/31 significant effects from the primary models, while four effects that were non-significant in the primary models reached statistical significance after applying this correction for missing data.

Finally, the results of follow-up models examining the effects of smartphone application use binned hourly relative to sleep onset on both sleep duration and step count are presented in **Supplementary Figure S1**. Clear temporal moderation effects were evident for both sleep duration and step count. Zero significant associations between smartphone use and total sleep duration were observed for any hourly time use bin in the 12 hours prior to sleep onset, ($\beta = [-12.05 - 4.17$ s], $P = [.051 - 0.994]$), while smartphone application use recorded in the hour immediately following sleep onset and in each hourly bin between four and ten hours after initiating sleep all predicted significant, often large, reductions in sleep duration ($\beta = [-91.73 - -17.02$ s], $P = [< 0.001 - 0.013]$). Application use recorded between ten and 12 hours after sleep initiation were associated with greater sleep duration on the previous night ($\beta = [15.44 - 33.61$ s], $P = [< 0.001 - 0.019]$). In contrast, 7/9 hourly bins for smartphone application use recorded between 12

Table 3
Within/Between-Person and Combined Smartphone Use Effects on Sleep and Physical Activity.

Outcome	Combined Effect		Within-Person Effect				Between-Person Effect		
Minutes of Smartphone Application Use (6:00AM – 7:59PM)									
	β	β_{Std}	P	β	β_{Std}	P	β	β_{Std}	P
Bedtime (Min Delayed)	0.018	0.019	.348	0.005	0.005	.820	0.116	0.123	.038
Waketime (Min Delayed)	0.020	0.020	.395	0.011	0.011	.668	0.062	0.063	.276
Total Sleep (Min)	0.006	0.009	.718	0.020	0.030	.274	−0.043	−0.064	.209
Exercise (Min)	−0.046	−0.128	<0.001	−0.052	−0.144	<0.001	−0.004	−0.011	.811
Sedentary Behavior (Min)	0.169	0.069	<0.001	0.168	0.068	<0.001	0.369	0.151	.017
Total Steps	−7.105	−0.209	<0.001	−7.344	−0.216	<0.001	−4.103	−0.120	.037
Minutes of Smartphone Application Use (8:00PM – 9:59PM)									
		Std	P		Std	P		Std	P
Bedtime (Min Delayed)	0.082	0.023	.244	−0.002	−0.001	.979	0.925	0.257	<0.001
Waketime (Min Delayed)	0.118	0.031	.167	0.076	0.021	.406	0.387	0.105	.102
Total Sleep (Min)	−0.063	−0.024	.295	0.024	0.009	.712	−0.462	−0.179	.001
Exercise (Min)	−0.038	−0.028	.073	−0.043	−0.031	.058	0.003	0.002	.970
Sedentary Behavior (Min)	0.222	0.024	<0.001	0.218	0.023	<0.001	0.908	0.097	.153
Total Steps	−6.783	−0.052	<0.001	−6.846	−0.053	<0.001	−5.73	−0.044	.484
Minutes of Smartphone Application Use (10:00PM – 5:59AM)									
		Std	P		Std	P		Std	P
Bedtime (Min Delayed)	0.369	0.269	<0.001	0.330	0.240	<0.001	0.599	0.436	<0.001
Waketime (Min Delayed)	0.246	0.176	<0.001	0.190	0.136	<0.001	0.495	0.353	<0.001
Total Sleep (Min)	−0.158	−0.161	<0.001	−0.154	−0.158	<0.001	−0.170	−0.173	<0.001
Exercise (Min)	−0.008	−0.015	.356	−0.006	−0.011	.530	−0.022	−0.041	.361
Sedentary Behavior (Min)	0.067	0.019	.001	0.064	0.018	.002	0.322	0.091	.124
Total Steps	−3.218	−0.065	<0.001	−2.655	−0.054	.001	−9.204	−0.186	<0.001
Minutes of Keyboard Use (6:00AM – 7:59PM)									
		Std	P		Std	P		Std	P
Bedtime (Min Delayed)	0.196	0.030	.038	0.155	0.023	.129	0.451	0.068	.079
Waketime (Min Delayed)	−0.006	−0.001	.952	−0.039	−0.006	.742	0.129	0.019	.596
Total Sleep (Min)	−0.107	−0.023	.167	−0.101	−0.021	.258	−0.124	−0.026	.431
Exercise (Min)	−0.087	−0.034	.005	−0.116	−0.046	<0.001	0.059	0.023	.440
Sedentary Behavior (Min)	0.344	0.020	<0.001	0.329	0.019	<0.001	1.578	0.092	.027
Total Steps	−12.287	−0.052	<0.001	−12.95	−0.054	<0.001	−7.029	−0.030	.408
Minutes of Keyboard Use (8:00PM – 9:59PM)									
		Std	P		Std	P		Std	P
Bedtime (Min Delayed)	0.503	0.026	.045	0.375	0.019	.152	2.021	0.104	.028
Waketime (Min Delayed)	0.192	0.010	.501	0.080	0.004	.793	1.112	0.056	.207
Total Sleep (Min)	−0.220	−0.016	.298	−0.184	−0.013	.423	−0.444	−0.032	.441
Exercise (Min)	0.103	0.014	.211	0.062	0.008	.476	0.441	0.059	.081
Sedentary Behavior (Min)	0.217	0.004	.284	0.186	0.004	.360	4.071	0.081	.074
Total Steps	0.045	0.000	.995	−0.069	−0.000	.993	1.517	0.002	.957
Minutes of Keyboard Use (10:00PM – 5:59AM)									
		Std	P		Std	P		Std	P
Bedtime (Min Delayed)	1.548	0.134	<0.001	1.356	0.118	<0.001	3.937	0.341	<0.001
Waketime (Min Delayed)	0.942	0.080	<0.001	0.755	0.064	<0.001	2.588	0.219	<0.001
Total Sleep (Min)	−0.707	−0.086	<0.001	−0.661	−0.080	<0.001	−1.017	−0.124	.005
Exercise (Min)	0.065	0.014	.220	0.064	0.015	.248	0.080	0.018	.648
Sedentary Behavior (Min)	−0.097	−0.003	.446	−0.113	−0.004	.374	2.991	0.100	.069
Total Steps	0.142	0.000	.976	2.371	0.006	.626	−41.43	−0.100	.038

Smartphone use effects from linear mixed effects models with individual random intercepts. Covariates varied by model and included sex, age, weekend, school night, summer, Fitbit wear time, and OS. β are the effect of one additional minute of smartphone use and β_{std} are standardized effects. Between-person effects represent participants' mean smartphone use, within-person effects are daily deviations from the participants' mean use, and combined effects do not disaggregate within or between-person effects.

and three hours before sleep onset were associated with significant step count reductions (β =[−27.96 – −8.78 steps], P =[<0.001 – 0.036]) while *zero* significant associations were observed in the three hours prior to sleep onset or after initiating sleep (β =[−11.02 – −4.02 steps], P =[.067 – 0.866]).

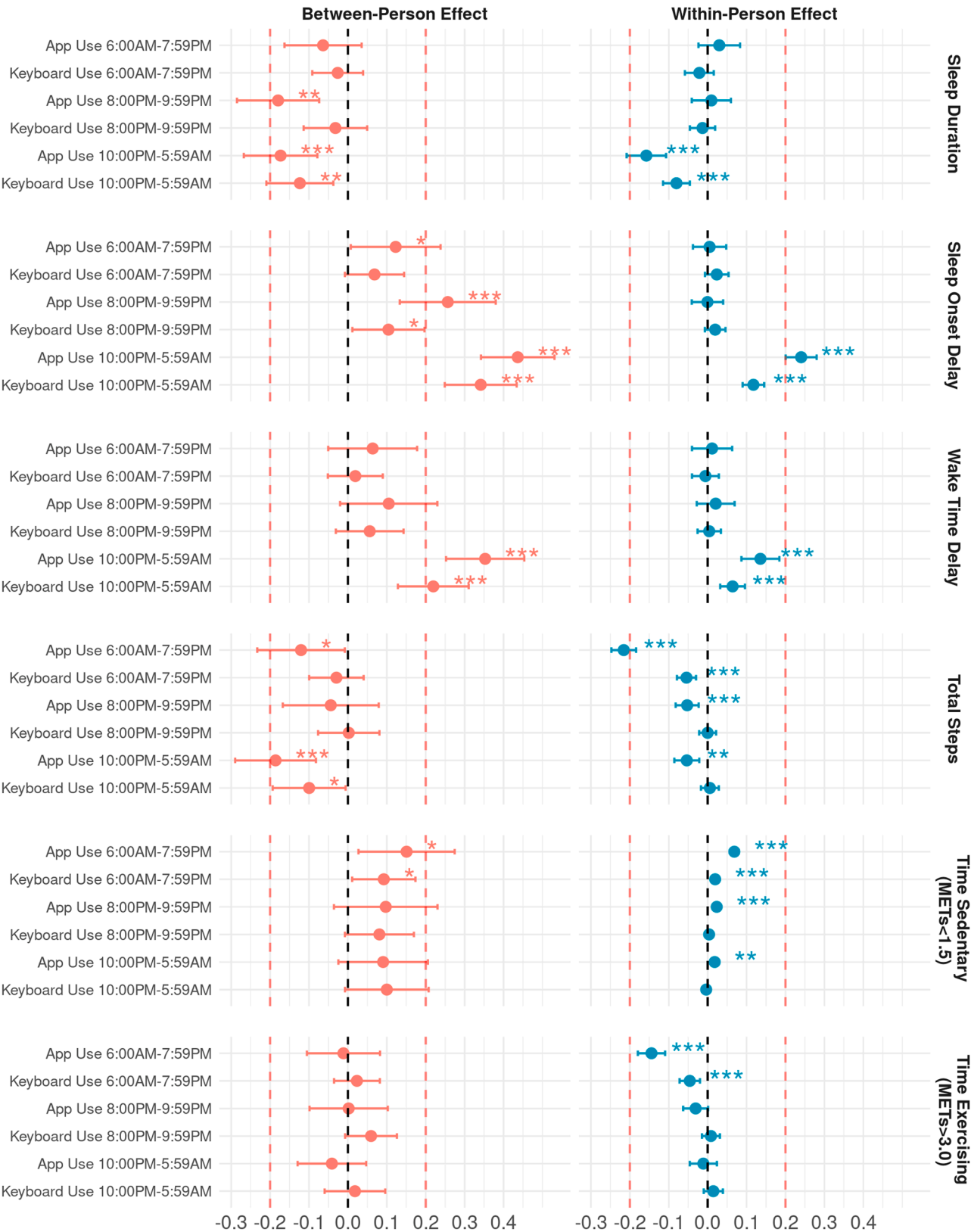
4. Discussion

The present study evaluated relationships between adolescent smartphone use, sleep, and physical activity using three weeks of continuous passive sensor data. The study aimed to (1) quantify associations between these behaviors using passive sensor measures (2) investigate whether their relationships differed by time of day, and (3) determine the extent to which relationships reflected differences in mean smartphone use or daily variations in individuals' use.

The study benefits from a well-powered, sociodemographically

diverse sample. Furthermore, to the authors' knowledge, no prior study has reported associations between concurrently collected passive sensor measures of adolescent smartphone use, physical activity, and sleep. The study's continuous, intensive-longitudinal design also facilitated analyses unavailable in cross-sectional or even daily-level data, including consideration of differences in smartphone use effects by time of day and analyses of within-person and between-person effects.

All six sleep and activity measures were significantly related to at least two smartphone use measures, corroborating previously reported self-report-based associations between greater adolescent smartphone use, reduced physical activity, and reduced and delayed sleep (Brautsch et al., 2023; Xiao et al., 2022). Some effects, like the association between late-night application use and sedentary behavior, were statistically significant but substantively negligible (e.g. Standardized β <0.05). Others, like the effect of daytime smartphone use on step counts, were small but behaviorally meaningful (Standardized β ≈0.20). The largest



observed associations, like late night smartphone use and sleep onset time, were moderate in magnitude (Standardized $\beta \approx 0.30$).

We further observed differences in associations by time of day. Daytime/evening smartphone use measures were more strongly associated with physical activity, predicting small, statistically significant effects on all three physical activity measures. Contrastingly, late-night smartphone use appeared more predictive of sleep outcomes, with significant, small-moderate associations with all three sleep measures. This may reflect the particularly disruptive effects of late-night smartphone use on sleep, possibly through increased arousal, blue-light exposure, or youth interrupting sleep to use their devices during the night (LeBourgeois et al., 2017). This latter explanation appears especially consistent with results from a supplementary analysis examining smartphone use effects binned hourly for the 12 hours before and after sleep onset, which found that smartphone application use effects on sleep duration were small and non-significant prior to participants initiating sleep, while overnight use recorded after initial sleep onset was associated with large reductions in sleep duration.

Alternatively, as smartphone use is largely stationary (Fennell et al., 2019) and cannot occur while participants sleep, associations may simply reflect that smartphone use cannot occur simultaneously with activities like sleep or exercise. Smartphone use may thus reduce sleep and physical activity by competing for adolescents' time, consistent with the time displacement hypothesis (Lizandra et al., 2019). However, as causality cannot be readily inferred from these associational results, such interpretations should be made cautiously. Even so, these findings do plausibly indicate potential health risks of excessive smartphone use, such that adolescents may be advised against using their phones late at night to prevent sleep disruptions.

Disaggregating smartphone use into within-person and between-person effects frequently yielded significant effects of both average smartphone use and of daily variations in use. Significant within-person effects of at least two smartphone use measures were observed for all six physical activity and sleep outcomes while at least two significant between-person effects were observed for every outcome except for minutes of exercise. This stands in contrast to Burnell et al. (2022), who found that relationships between self-reported smartphone use and digitally-recorded sleep were largely between-person effects (Burnell et al., 2022). This discrepancy may reflect differences between self-reported and sensor-based measures of smartphone use or the present study's separation of daytime/evening and late-night smartphone use (Alexander et al., 2024).

These within-person associations suggest that on days when adolescents deviated from their typical smartphone use patterns, their sleep patterns and activity levels also changed, potentially consistent with hypothesized causal effects of smartphone use (LeBourgeois et al., 2017; Lizandra et al., 2019). The study's naturalistic observational design suggests high ecological validity and its within-subject analyses offer a stronger evaluation of causality than much prior observational research: holding constant individual differences which often confound associations observed between-subjects. However, results may also reflect reverse-causality (e.g. a lack of opportunities for physical activity drive adolescents toward other behaviors, like smartphone use) or the effect of some within-person, time-varying confounder, like daily fluctuations in mood that impact both smartphone use and sleep or physical activity (Rutter, 2001). Further replication of these relationships across alternative causally informative designs, like randomized controlled experiments, (see, for example, Pedersen et al., 2022 or Pickard et al., 2024) would offer strong evidence for causal effects of excessive smartphone use on adolescent health, allowing for triangulation across causally informative methods with complementary strengths and limitations (Pedersen et al., 2022; Pickard et al., 2024; Taylor and Asmundson, 2008). Nonetheless, the present study suggests that, on balance, modifying smartphone behaviors plausibly contributes to changes in physical activity and/or sleep.

In addition to limitations on causal inference, further constraints

impact this study's interpretation. Recruitment and participation in the ABCD wearables trial differed across sociodemographic backgrounds, possibly impacting the study's generalizability to individuals from under sampled groups (Kim et al., 2023). The study considered device usage time across all application categories and without consideration for content. It thus did not investigate how associations with sleep or physical activity may differ across different phone-based activities, like gaming, video streaming, or social media use. Investigating effects of specific smartphone use content, in addition to total quantity, is an important future direction to be addressed using passive sensor data. Though this study benefits from use of validated sensor-based measures (Alexander et al., 2024; Godino et al., 2020), sources of errors from device issues (e.g., not wearing the Fitbit for bursts of time; memory capacity on a smartphone) may limit full sensitivity of measures. Keyboard data collection procedures differed between Android and iOS device users and application use was only available for Android devices. As an estimated 87 % of teens in the United States own an iPhone, (Wolfmeyer et al., 2024) including iOS device users substantially strengthens the study's representativeness and ecological validity. Nonetheless, while EARS smartphone measures are validated in both Android and iOS devices (Alexander et al., 2024) and OS differences were statistically controlled for via a fixed effects covariate, differences in collection procedures across OS are nonetheless an important source of measurement variability for keyboard use, which reduces the comparability of findings across smartphone operating systems. Additionally, Fitbit-based measures of sleep and physical activity are subject to greater measurement error than "gold-standard" sensor measures used in clinical research settings, especially for our measures of time spent sedentary or exercising, which are known to be error prone (Feehan et al., 2018; Germini et al., 2022). Screen use measurement was limited to participants' personal smartphones and did not include their use of other screen-based media or, possibly, use of others' smartphone devices. Variation in smartphone use was greater at the within-person than at the between-person level, which reduced statistical power and precision when estimating between-person effects relative to within-person effects. Lastly, as is common in sensor-based intensive longitudinal studies, (Seizer et al., 2024) data were subject to substantial missingness. While results remained highly similar after propensity-weighting to correct for selection biases, missing data were assumed at least MAR and thus unmeasured influences on missingness may bias the study's results (Seaman and White, 2013).

5. Conclusion

This passive-sensing-based trial offers further evidence for relationships between increased adolescent smartphone use, sleep disruptions, and reduced physical activity. Relationships largely persisted within and between subjects and differed by time of day, with stronger relationships between physical activity and daytime/evening use and between sleep and late-night use. Findings highlight the strengths of passive sensor methods, including their ability to collect informative, intensive longitudinal data without reporting biases. They further indicate the need for causally informative research on the impacts of adolescent smartphone use, which may inform guidelines on responsible adolescent smartphone use and aid the development of interventions to mitigate potential harms. While awaiting further study, youth may be advised of the likely negative relationship between large amounts of smartphone use, particularly late in the night, on sleep and physical activity.

CRedit authorship contribution statement

Jordan D. Alexander: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tam T. Nguyen-Louie:** Writing – review & editing, Conceptualization. **Suhaani Gupta:** Writing – review & editing. **Kevin M. Cummins:** Writing – review & editing,

Methodology. **Natasha E. Wade:** Writing – review & editing, Supervision, Data curation, Conceptualization.

Declaration of competing interest

None of the authors have any potential conflicts of interest, either real or perceived, to disclose.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.psychres.2025.116523](https://doi.org/10.1016/j.psychres.2025.116523).

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